

# Introduction to Quadratic Functions

## Objectives:

- Identify quadratic functions
- Determine how linear and quadratic functions are different.
- Sketch parabolas in standard form by using the vertex formula.

## Common Core Standards:

F.IF.1 - Understand that a function from one set (called the domain) to another set (called the range) assigns to each element of the domain exactly one element of the range.

F.IF.4 - For a function that models a relationship between two quantities, interpret key features of graphs and tables in terms of the quantities, and sketch graphs showing key features given a verbal description of the relationship.

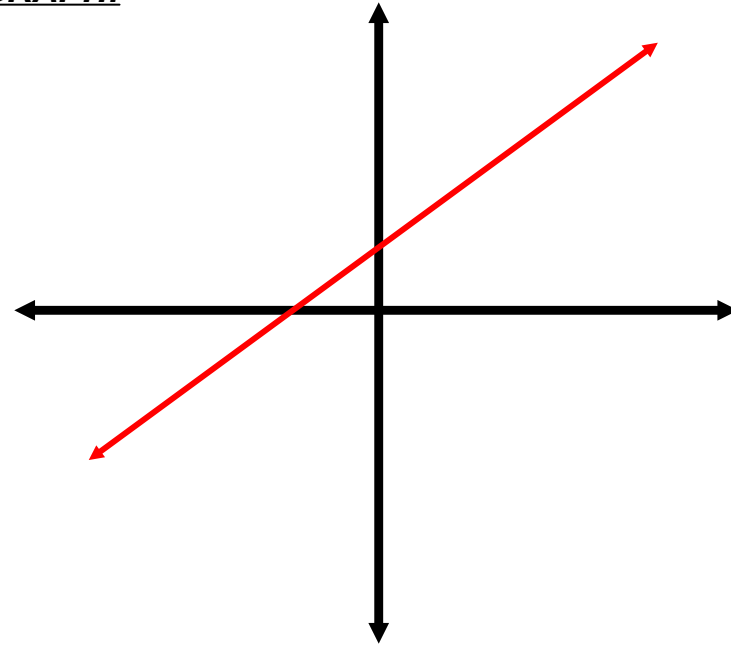
F.IF.7a - Graph functions linear and quadratic functions and show intercepts, maxima, and minima.

A.REI.10 - Understand that the graph of an equation in two variables is the set of all its solutions plotted in the coordinate plane. Often forming a curve (which could be a line).

## Linear Function

- Line
- First Degree
- Example:  $y = 4x + 5$

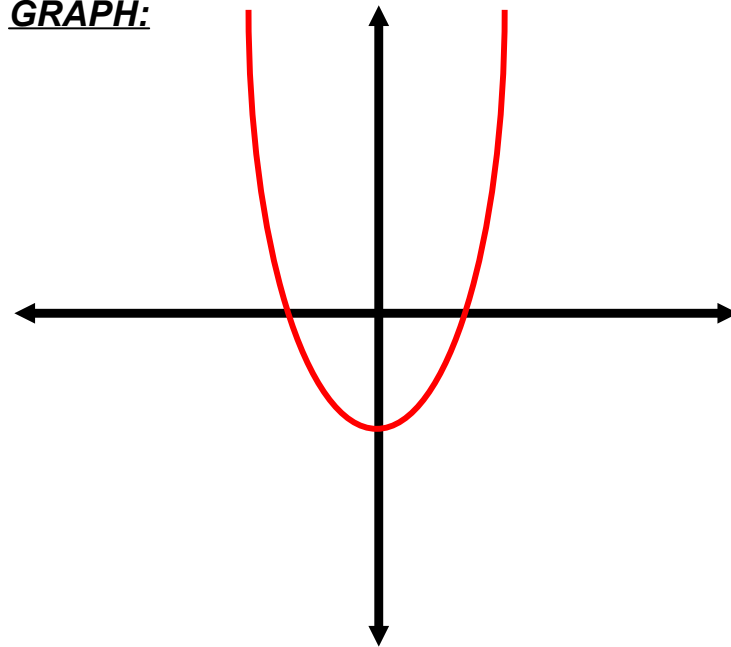
GRAPH:



## Quadratic Function

- Parabola
- Second Degree
- Examples:  $y = x^2 + 4x + 1$   
 $y = x^2$   
 $y = x^2 + 3$

GRAPH:



## Standard Form of a Quadratic Function

$$y = Ax^2 + Bx + C$$

A is the coefficient of  $x^2$

B is the coefficient of  $x$

C is the constant

Identify A, B, and C for the following quadratic functions.

$$y = 3x^2 + 5x + 2$$

$$A = 3$$

$$B = 5$$

$$C = 2$$

$$y = 6x - 7x^2$$

$$A = -7$$

$$B = 6$$

$$C = 0$$

## Vertex of a Parabola

The vertex will be the starting point when you sketch a parabola.

Vertex Formula:  $x = \frac{-b}{2a}$

The Vertex is a Point: (x, y)

$$x = \frac{-b}{2a} \quad y = \text{Plug } x \text{ into the equation!}$$

Find the vertex of  $y = 2x^2 + 8x + 4$ .

$$A = 2, B = 8, C = 4$$

Find x:

$$x = \frac{-8}{2(2)}$$

$$x = \frac{-8}{4}$$

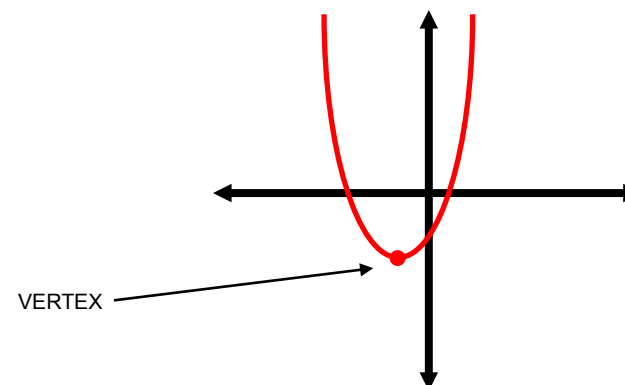
$$x = -2$$

Find y:

$$y = 2(-2)^2 + 8(-2) + 4$$

$$y = 8 - 16 + 4$$

$$y = -4$$



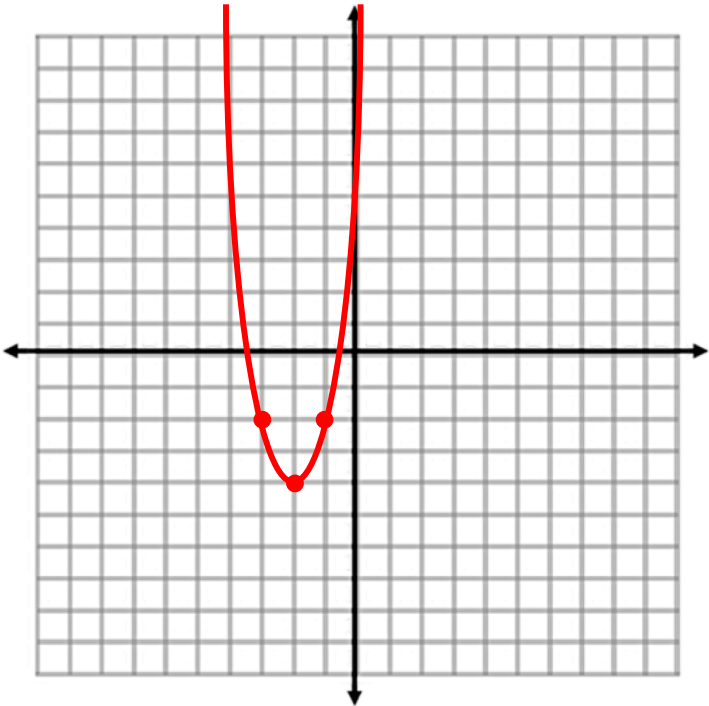
**Vertex: (-2, -4)**

**Graph  $y = 2x^2 + 8x + 4$  on the coordinate plane.**

- Find and graph the vertex.
- Use an x|y chart to find a point on each side of the vertex. (Pick an x-value on each side.)
- Use the points to sketch a parabola.

**Vertex: (-2, -4)**

| x  | $y = 2x^2 + 8x + 4$       | y  | (x, y)                           |
|----|---------------------------|----|----------------------------------|
| -1 | $y = 2(-1)^2 + 8(-1) + 4$ | -2 | (-1, -2)<br><small>LEFT</small>  |
| -3 | $y = 2(-3)^2 + 8(-3) + 4$ | -2 | (-3, -2)<br><small>RIGHT</small> |



*The vertex is the lowest point on the graph, so it is called the minimum.*

Graph  $y = -3x^2$  on the coordinate plane.

- Find and graph the vertex.
- Use an x|y chart to find a point on each side of the vertex. (Pick an x-value on each side.)
- Use the points to sketch a parabola.

Vertex:  $A = -3, B = 0, C = 0$

Find x:

$$x = \frac{-0}{2(-3)}$$

$$x = 0$$

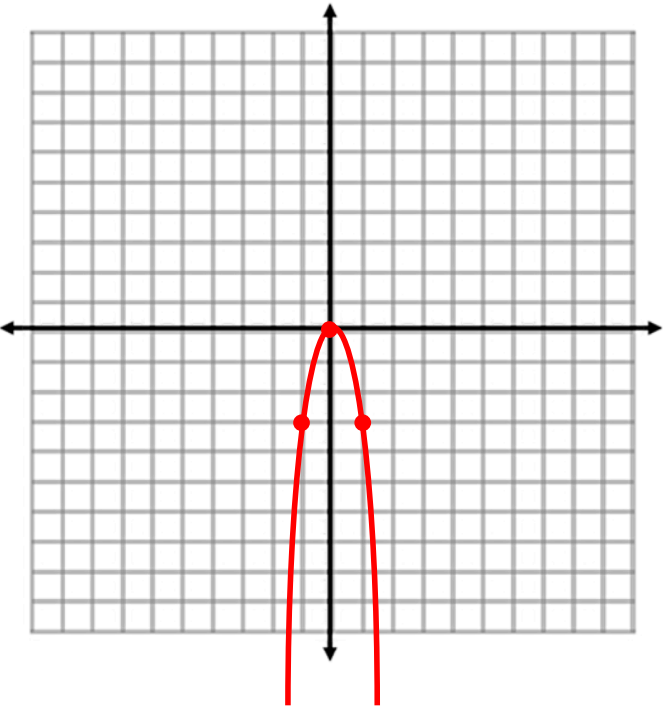
Find y:

$$y = -3(0)^2$$

$$y = 0$$

Vertex: (0, 0)

| x  | $y = -3x^2$    | y  | (x, y)                          |
|----|----------------|----|---------------------------------|
| -1 | $y = -3(-1)^2$ | -3 | (-1, -3)<br><small>LEFT</small> |
| 1  | $y = -3(1)^2$  | -3 | (1, -3)<br><small>RIGHT</small> |



The vertex is the highest point on the graph, so it is called the maximum.

Graph  $y = x^2 + 2x$  on the coordinate plane.

- Find and graph the vertex.
- Use an x|y chart to find a point on each side of the vertex. (Pick an x-value on each side.)
- Use the points to sketch a parabola.

Vertex:  $A = 1, B = 2, C = 0$

Find x:

$$x = \frac{-2}{2(1)}$$

$$x = -1$$

Find y:

$$y = (-1)^2 + 2(-1)$$

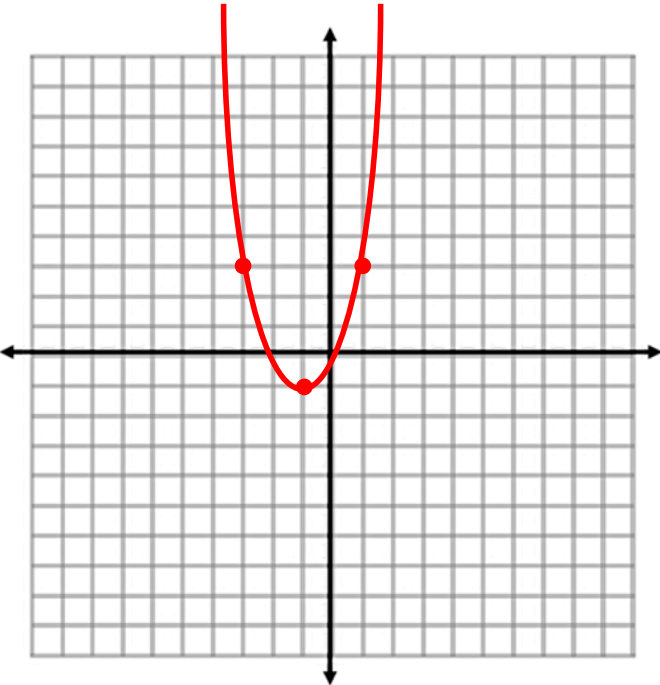
$$y = -1$$

Vertex:  $(-1, -1)$

| x  | $y = x^2 + 2x$       | y | (x, y)    |
|----|----------------------|---|-----------|
| -3 | $y = (-3)^2 + 2(-3)$ | 3 | $(-3, 3)$ |
| 1  | $y = (1)^2 + 2(1)$   | 3 | $(1, 3)$  |

LEFT

RIGHT



The vertex is the lowest point on the graph, so it is called the minimum.